

1. A clearly stated answer with specific numbers of pizzas and sandwiches to be produced.

The most you can make within the time frame of 60 minutes is 6 pizzas and 4 sandwiches to maximize profit while having 10 items. The maximum profit from this combination is \$210.

2. An objective function

Each pizza is \$25 in profit

Each sandwich earns \$15 profit

$$c = (25, 15)$$

$$f(x) = 25x_1 + 15x_2$$

3. Constraint inequalities

x_1 = amount of pizzas

x_2 = amount of sandwiches

Amount Constraints:

The number of pizzas and sandwiches must be 10.

$$x_1 + x_2 = 10$$

Time Constraints

The time it takes to make the pizzas and sandwiches must take less than 60 minutes

$$8x_1 + 3x_2 \leq 60$$

4. A chart or table of the setup, like the one on slide 11

The chart should have the profit at the end, and it should have another column representing the number of items and how many items can be made (10) as well as the amount of time available

	Pizza	Sando	Available
Mins	8	3	60
Items	1	1	10
Cost	\$25	\$15	

5 & 6 All work necessary to determine the final answer including solutions for three options (only pizzas; only sandwiches; and the balanced option) .

All necessary algebra to determine the balanced option as well as the solo options

Student includes calculations of pizzas only, sandwiches only, AND the balanced option.

Student picks the one with the greatest profit.

Balanced Solution has the greatest profit

Initially:

$$8x_1 + 3x_2 \leq 60$$

$$x_1 + x_2 \leq 10$$

Isolating x_2 in the second equation:

$$x_2 = 10 - x_1$$

Replacing x_2 in the first equation:

$$8x_1 + 3(10 - x_1) = 60$$

Multiplying 3 times the x_2 replacement:

$$8x_1 + 30 - 3x_1 = 60$$

Combining like terms:

$$5x_1 + 30 = 60$$

Isolating x_1 subtract 30 from both sides:

$$5x_1 = 30$$

Divide by 5

$$x_1 = 6$$

Replacing x_1 in the second equation:

$$6 + x_2 = 10$$

$$x_2 = 4$$

$$x_1 = 6 \text{ pizzas}$$

$$x_2 = 4 \text{ sandwiches}$$

$$\text{Profit} = (25 * 6 \text{ Pizzas}) + (15 * 4 \text{ Sandwiches}) = 150 + 60 = \$210$$

Solo Pizza in 60 mins

Initially:

$$8x_1 + 3x_2 \leq 60$$

$$x_1 + x_2 \leq 10$$

$$x_1 = \text{amount of pizzas}$$

Set sandwich to zero

$$x_2 = 0$$

Sub x_2 in the first equation:

$$8x_1 + 3(0) \leq 60$$

$$8x_1 \leq 60$$

Divide by 8

$$8/8x_1 = 60/8$$

$$x_1 = 7.5$$

$$x_1 = 7.5 \text{ rounded down to } 7$$

$$7 \leq 10$$

$$x_1 = 7$$

Can make 7 pizzas in an hour

$$\text{Profit} = 7 \text{ pizzas} * \$25 = \$175.$$

Solo Sandwiches in 60 mins

Initially:

$$8x_1 + 3x_2 \leq 60$$

$$x_1 + x_2 \leq 10$$

x_2 = amount of sandwiches

Set pizza to zero

$$x_1 = 0$$

Sub x_1 in the first equation:

$$8(0) + 3x_2 \leq 60$$

$$3x_2 \leq 60$$

Divide by 3

$$x_2 \leq 20$$

Replacing x_1 in the second equation:

$$0 + x_2 \leq 10$$

$x_2 = 10$ is less than 20 so we can say ten.

Can make well over 10 sandwiches in an hour. But with our amount constraint, 10 is the limit.

Profit = 10 sandwiches * 15 = \$150